

Numerical Simulation of Learning Dynamics in Multimodal RL

A tunable diversity correction for outcome-level mode collapse

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CSE 402 Numerical Analysis and Simulation Sessional

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Background

Base Paper: The Collapse Mechanism

Sinha, Elango and Liu (2026), *arXiv:2601.21669*.

Many RL tasks have **several equally good answers**: valid reasoning chains, distinct drug candidates, different correct programs. We want the policy to keep all of them, not pick one.

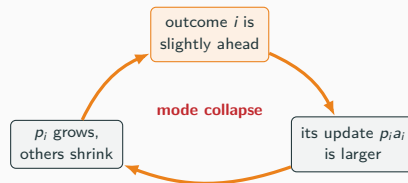
The standard objective is expected return

$$J(\theta) = \mathbb{E}_{o \sim p_\theta} [r(o)] = \sum_i p_i r_i.$$

Under gradient flow the paper proves (Thm. 3.1)

$$\dot{z}_i = p_i a_i, \quad a_i = r_i - \bar{r}.$$

An outcome's update is scaled by **how likely it already is**. That is a rich-get-richer loop, and it runs even when two outcomes have identical reward.



Collapse is a property of **the objective itself**. Entropy bonuses and KL penalties push against it, but they do not remove the cause.

Base Paper: Inverse Probability Scaling

Inverse Probability Scaling. Divide the reward by the probability of the outcome that produced it:

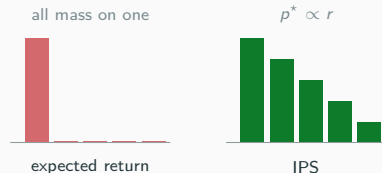
$$\tilde{r}(o) = \frac{r(o)}{p_{\theta}(o)}.$$

The $1/p$ cancels the p_i multiplier, so (Thm. 4.1, Cor. 4.2)

$$\dot{z}_i = r_i - p_i \sum_k r_k \implies p_i^* = \frac{r_i}{\sum_k r_k}.$$

In practice $p_{\theta}(o)$ is unknown. IPS-GRPO estimates it from a group of G sampled rollouts and clips (their Eq. 9):

$$\tilde{r}(o_g) = \frac{r(o_g)}{\max\{\hat{p}(o_g), \varepsilon\}}.$$



Left open by the paper:

1. Only two settings exist: plain (collapse) or exact proportionality.
2. The group size G and clip ε are ablated empirically only (their Table 4).
3. Their own limitation section: finite-group estimates “can be noisy, requiring probability clipping that introduces a bias–variance trade-off”. No analysis is given.

Project Overview

Problem Statement and Extension

The paper gives one fixed correction. We asked three questions it does not answer.

1. Can the correction be tuned?

Replace r/p by r/p^α . Redoing the paper's own gradient calculation gives

$$\dot{z}_i = r_i p_i^{1-\alpha} - p_i \sum_k r_k p_k^{1-\alpha}, \quad \boxed{p_i^* \propto r_i^{1/\alpha}}$$

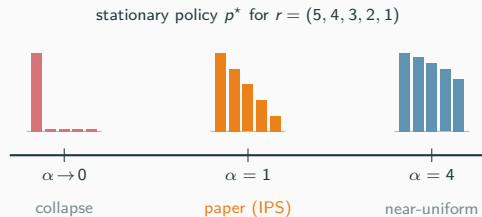
One knob spans collapse, the paper's result, and near-uniform.

2. What does a finite group cost?

Derive the effect of G and ε instead of ablating it.

3. How wrong is the estimate $\hat{p}$?

The error analysis the paper lists as an open limitation.



α becomes a **diversity dial**: the same algorithm, one extra exponent.

System Model

An **outcome-selection bandit**: K outcomes, fixed rewards $r_i > 0$, policy stored as logits and squashed by softmax,

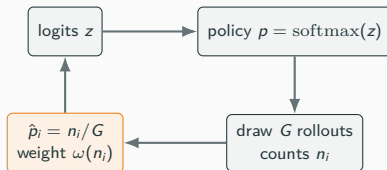
$$p_i = \frac{e^{z_i}}{\sum_k e^{z_k}}.$$

Nothing else is in the model. No network, no exploration schedule, no critic. So any collapse we see has **exactly one possible cause**, which is the point the paper argues.

Two objects are studied throughout:

- the **idealised flow** $\dot{z} = f(z)$, an ODE in K variables;
- the **sampled update**, one noisy step built from G draws.

Parts 1–4 use the exact p here



Part 5 lives here

$$\hat{g}_i = \hat{p}_i r_i \omega(n_i) - p_i \sum_k \hat{p}_k r_k \omega(n_k), \quad z \leftarrow z + h \hat{g}$$

Work Breakdown

Part	Focus	Key content
1	Collapse baseline, $\alpha = 0$	Integrate the flow with Euler and RK4, verify orders, show the exact tie is provably flat, and that sampling noise alone still collapses it
2	The α knob	Stationary law $p^* \propto r^{1/\alpha}$, linearisation and eigenvalue spectrum, convergence rates, explicit-method stability limits, first finite- G ceiling
3	Solving $\dot{z} = 0$ directly	Bisection, secant and Newton on three equivalent formulations; convergence orders, basins, conditioning; nested solver for the finite- G fixed point
4	Multi-parameter atlas	19,125 stationary points over $(\alpha, G, \varepsilon, \rho)$, support and entropy for general K , and whether interpolation can replace solving
5	The estimator itself	Bias and variance of the clipped weight, Richardson extrapolation, golden-section search, power and QR methods, moving averages

One shared library (src/), one script per part (experiments/), one write-up per part, and 68 unit tests.

Part 1: Collapse Baseline

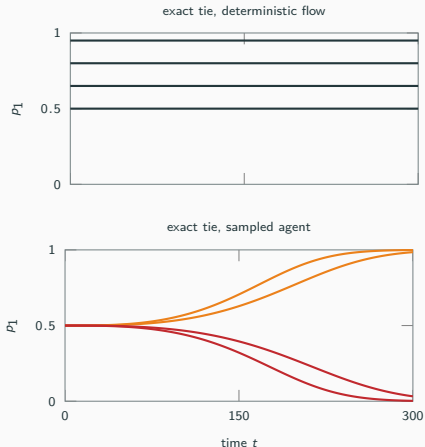
Part 1: Objective

Before correcting collapse we had to reproduce it, and to be precise about a claim the paper states informally.

The paper says collapse happens “even with exactly equal rewards”. For the **deterministic** flow that is not literally true: if all r_i are equal then $a_i \equiv 0$, so $\dot{z} \equiv 0$ and nothing moves at all.

What actually drives collapse under a tie is **sampling noise**. A finite group never sees the exact mean reward, so the realised advantage is never zero, and the p_i multiplier then amplifies it.

So Part 1 runs both and separates them.



Same rewards, same starting policy. Only the noise differs.

Part 1: Numerical Methods

Solution of ODEs. The flow is a coupled system in K unknowns,

$$\dot{z}_i = r_i p_i - p_i \sum_k r_k p_k.$$

Integrated with Euler $z_{n+1} = z_n + hf(z_n)$ and classical RK4

$$z_{n+1} = z_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4).$$

Truncation and round-off error. Empirical order from a step-size sweep,

$$\text{order} = \text{slope of } \log E \text{ vs } \log h.$$

Below $h \approx 0.025$ the RK4 error stops falling: the truncation term has dropped under the round-off floor.

Least-squares regression. Collapse time against reward gap, fitted in log-log space,

$$\log t_{\text{col}} = m \log \Delta + c \Rightarrow t_{\text{col}} \approx 53.94 \Delta^{-1.00}.$$

The same fit gives the group-size law $t_{\text{col}} \propto G^{1.07}$.

Random number generation. The sampled agent draws outcomes by **inverse transform sampling**: build the CDF of p , draw $u \sim U(0, 1)$, take

$$o = \min\{i : \sum_{k \leq i} p_k \geq u\}.$$

Visualisation. Semi-log and log-log axes, since both the transition and the fitted laws span decades.

Part 1: Results and Analysis

The tie is provably flat. Four different starting policies, $T = 300$:

$$\max_t |\Delta p_1| = 1.11 \times 10^{-16}$$

which is machine epsilon, not a slow drift. This **sharpens** the paper's claim rather than contradicting it.

Noise alone collapses it. From an exactly symmetric start ($z_0 = 0$), $K = 5$, 200 seeds:

outcome	O1	O2	O3	O4	O5
won	18.5%	22.5%	19.0%	19.0%	21.0%

All inside the 95% band $[14.5, 25.5]\%$ around $1/K$. The winner is arbitrary.

Two clean scaling laws.

$$t_{\text{col}} \approx 53.94 \Delta^{-1.00}, \quad R^2 = 0.99999994$$

$$t_{\text{col}} \propto G^{1.07}, \quad R^2 = 0.974$$

A bigger group delays collapse roughly linearly. That is a statement about a real GRPO hyperparameter.

Integrators behave.

	order	error at $h = 0.05$
Euler	1.00	1.4×10^{-4}
RK4	3.99	5.1×10^{-8}

Reported honestly: we looked for an Euler instability here and found none. The right-hand side is bounded and self-damping at $\alpha = 0$, so no step size can overshoot. Part 2 is where instability appears.

Part 2: The Diversity Exponent

Part 2: Objective

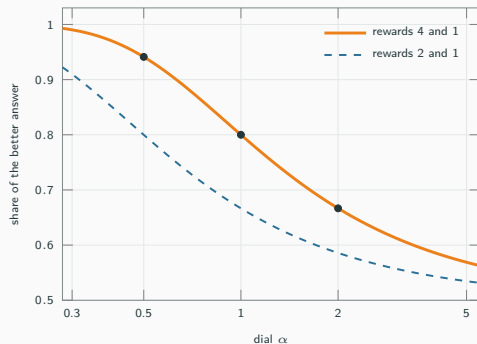
The idea. The paper divides each reward by p . We divide by p^α instead, so α becomes a **dial**:

- $\alpha = 0$: no fix, one answer takes everything.
- $\alpha = 1$: the paper's fix, share matches reward.
- large α : shares become almost equal.

Where training ends up. Training stops when every answer earns the same corrected reward r_i/p_i^α . That gives $p_i^* \propto r_i^{1/\alpha}$.

Our four questions.

1. Does training really end up there?
2. How fast does it get there?
3. How big a step can the solver take?
4. What changes with a small sample group?



Dots: our predicted splits 94/6, 80/20, 67/33. The simulation lands on the curve to machine precision (7.5×10^{-15}).

Part 2: Numerical Methods (1/2)

Differential equations. Training is an ODE, $\dot{z} = f(z)$. Setting $\dot{z} = 0$ gives the end point $p^* \propto r^{1/\alpha}$.

Eigenvalues: how fast training settles. Near the end point the error shrinks like $e^{-\lambda t}$. The rates λ are the eigenvalues of one matrix, the Jacobian:

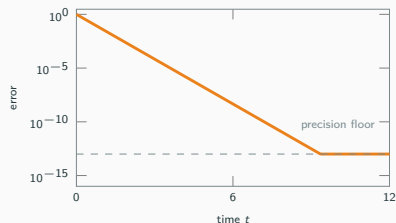
$$J^* = -\alpha \|r\|_{1/\alpha} (\text{diag } p^* - p^* p^{*\top}).$$

Big λ means fast, small λ means slow.

One eigenvalue is always 0: adding the same number to every logit changes nothing, so nothing pulls back in that direction.

Least squares: measuring the rate. On a log scale the error is a straight line. Its slope is the rate. We fit the slope and compare it with the eigenvalue.

Round-off error: where to fit. Only between 10^{-6} and 10^{-12} . Above that the line is not straight yet; below it the computer's precision floor takes over.



Part 2: Numerical Methods (2/2)

Stability: how big a step? Each solver step multiplies the error by some factor. If that factor is bigger than 1, the error grows every step and the run blows up. Safe steps:

$$h\lambda_{\max} < 2 \text{ (Euler)}, \quad h\lambda_{\max} < 2.785 \text{ (RK4)}.$$

Root finding: bisection. The RK4 limit 2.785 is the root of $x^3 - 4x^2 + 12x - 24 = 0$. We computed it by **bisection** instead of looking it up.

Then we used bisection on the real simulation to find the largest step that still settles. It matches the theory to within 0.03%.

Random sampling: real training. Real training does not know p . It draws a group of G samples and counts them:

$$G\hat{p}_i \sim \text{Binomial}(G, p_i).$$

Averaging over many groups gives an **effective weight** $w_G(p)$ that takes the place of $p^{-\alpha}$. The end point then solves

$$r_i w_G(p_i) = S,$$

again by bisection.

Checking the model. Every random simulation is compared against this average prediction.

Part 2: Results and Analysis

1. The dial works, and we can predict the speed.

Measured speed against the eigenvalue: within 0.002% over 27 runs.

2. Both ends of the dial are hard to simulate.

α	slowest rate	fastest rate	fast / slow
0.1	5.8×10^{-8}	0.089	1.5×10^6
1	1.15	4.53	3.9
5	7.2×10^3	9.3×10^3	1.3

Small α : the rates differ by a million times (a stiff problem). Large α : every rate is huge, so the step must be tiny. Part 1's step $h = 0.05$ blows up above $\alpha = 2.06$.

3. At $\alpha = 0$ there is no rate at all. The losing answer fades like $1/t$ (measured slope -1.007), not exponentially.

4. The key result: a small group caps the boost. A group of G samples sees a rare answer at most once, so it never estimates p below $1/G$. The boost $1/p^\alpha$ is capped:

$$\text{boost} \leq \min\{G, 1/\varepsilon\}^\alpha.$$

The weaker answer survives only if the cap beats the reward ratio:

$$\alpha > \frac{\ln(r_1/r_2)}{\ln \min\{G, 1/\varepsilon\}}$$

G	4	8	16	32	64
α needed (rewards 4, 1)	1.0	0.667	0.5	0.4	0.333

With $G = 4$ the paper's $\alpha = 1$ is not enough: the weaker answer dies (better answer's share 0.9965, not 0.8). Simulations match the prediction to 0.013.

Part 3: Root Finding Methods

Part 3: Objective

Part 2 reached p^* by integrating for many e-folding times. Part 3 solves $\dot{z} = 0$ **directly** and checks the root against the analytic answer.

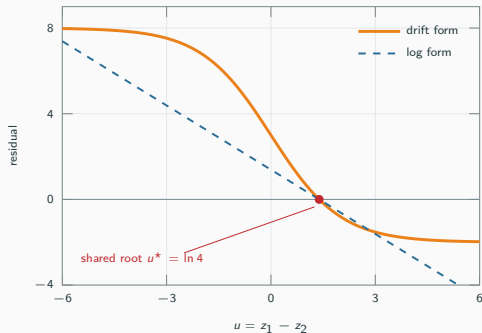
The same condition can be written three ways:

$$\text{drift: } F_i = \dot{z}_i$$

$$\text{balance: } F_i = r_i p_i^{-\alpha} - r_K p_K^{-\alpha}$$

$$\text{log: } F_i = \ln \frac{r_i}{r_K} - \alpha(z_i - z_K)$$

They share the same root. Dividing by p_i removes the very multiplier the paper blames for collapse, and it turns out to matter for Newton too.



The drift saturates at both ends, so a Newton step taken from far out lands even further away. The log form is a straight line, so Newton solves it in one step.

Part 3: Numerical Methods

Root finding. All three from the syllabus, on the same problem:

bisection: halve a sign-change bracket

$$\text{secant: } u_{n+1} = u_n - F_n \frac{u_n - u_{n-1}}{F_n - F_{n-1}}$$

$$\text{Newton: } u_{n+1} = u_n - F(u_n)/F'(u_n)$$

Measured order from the least-squares slope of $\ln e_{n+1}$ against $\ln e_n$.

Systems of equations. For $K > 2$, Newton solves

$$J(y) \delta y = -F(y)$$

by Gaussian elimination each iteration. J is the analytic Jacobian, not a finite difference.

Conditioning, and a gauge. Shifting every logit leaves $\hat{z}$ unchanged, so $J\mathbf{1} = 0$ and the full $K \times K$ system is **singular** (smallest singular value 7.1×10^{-17}). Fixing $z_K = 0$ removes it. At the root

$$J_{\text{balance}} = -\alpha \|r\|_{1/\alpha} I, \quad J_{\log} = -\alpha I,$$

so both have condition number exactly 1.

Nested root finding for finite G . Inner: invert the monotone $w_G(p_i) = S/r_i$ for each outcome. Outer: choose S with $\sum_i p_i(S) = 1$. Both levels are monotone with a guaranteed bracket, so bisection is safe and safeguarded Newton is fast.

Part 3: Results and Analysis

Textbook orders, confirmed. $K = 2$, $\alpha = 1$, root $u^* = \ln 4$:

method	iterations	order
bisection	56	linear (0.502)
secant	9	1.61
Newton (drift)	5	2.00
Newton (log)	1	exact

How you write the equation decides the basin.

Newton on the drift form converges from only **17%** of starts at $\alpha = 1$, but from 100% once $\alpha > 1.113$. On the balance form it converges from **100%** of starts at every α .

Backtracking does not rescue it: for $\alpha < 1$ every simplex vertex is a zero of the drift “at infinity”, and a line search is pulled into them.

Nested solver: Newton pays off. Over

$G \in \{4, 16, 64\} \times \alpha \in \{0.5, 1, 2\}$:

	outer iters	total work
bisection / bisection	43–55	4,506–13,769
Newton / Newton	6–16	225–1,703

Same answer to 9.8×10^{-14} , with 8 to 20 times less work.

A new result for $K > 2$. The two-outcome threshold understates the requirement. At $G = 16$, $r = (5, 4, 3, 2, 1)$:

α_c for	O2	O3	O4	O5
$K = 5$ mean field	0.0805	0.203	0.419	0.886
pairwise formula	0.0805	0.184	0.330	0.580

Some outcomes are lost at **every** α : here the weakest needs $G \geq 6$ before any exponent can keep it.

Part 4: Parameter Atlas

Part 4: Objective

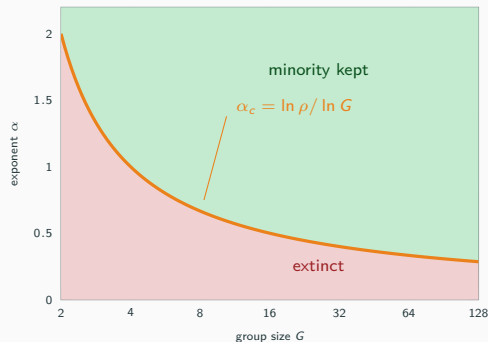
Parts 2 and 3 varied one or two parameters. Part 4 asks whether the survival boundary really holds **jointly** over all four, and whether a grid can stand in for solving.

The predicted boundary collapses to a single signed margin

$$m = \alpha \log \min\left\{G, \frac{1}{\varepsilon}\right\} - \log \rho,$$

with $\rho = r_1/r_2$. Survival should be exactly $m > 0$, independent of where in the four-dimensional box you are.

19,125 stationary points were solved to test that one inequality.



One slice ($\rho = 4$, $\varepsilon = 10^{-3}$) of a four-dimensional atlas.

Part 4: Numerical Methods

Interpolation. The atlas is a tensor-product grid in $(\log \alpha, \log \rho, \log \varepsilon)$. Inside one cell, with $t_d \in [0, 1]$ the fractional position along axis d , the **multilinear** interpolant is the weighted sum over the cell's 2^D corners:

$$\hat{f}(\mathbf{x}) = \sum_{\mathbf{c} \in \{0,1\}^D} f(\mathbf{x}_{\mathbf{c}}) \prod_{d=1}^D t_d^{c_d} (1 - t_d)^{1-c_d}.$$

This is the D -dimensional form of the linear interpolation in the syllabus, and it is exact at nodes and for any function affine in each coordinate separately.

Approximation error and order. Refine the grid and fit the slope of $\log(\text{RMSE})$ against $\log(\text{intervals per axis})$.

Root finding, vectorised. All 19,125 $K = 2$ solves run as one batched bisection with 64 deterministic halvings, and 40 of them are re-checked against the original scalar solver.

Model validation and verification. The script refuses to write a summary unless every check passes: phase classification, stationarity residual, normalisation, node exactness, and monotone refinement.

Where the theory says interpolation must degrade. The response is continuous but has derivative kinks at

$$m = 0 \text{ (extinction)}, \quad \varepsilon = 1/G \text{ (clip activates)},$$

so cells crossing those surfaces cannot keep second-order behaviour. We measure the two regions separately instead of quoting one rate.

Part 4: Results and Analysis

The boundary holds everywhere. Over $51 \times 15 \times 5 \times 5 = 19,125$ points:

$$\text{misclassifications} = 0/19,125$$

with worst stationarity residual 1.18×10^{-15} .

Two separate ceilings. ε can bind before G does. At $\alpha = 1$, $\rho = 4$:

p_2 at	$G=6$	$G=16$	$G=64$	$G=256$
$\varepsilon = 10^{-3}$	0.115	0.195	0.200	0.200
$\varepsilon = 0.3$	0	0	0	0

With $\varepsilon = 0.3$ the clip caps the boost at $3.33 < 4$, so no group size whatsoever rescues the minority.

Group size must grow with the problem. Smallest G that keeps every outcome at $\alpha = 1$:

K	spread 1.25	spread 32
2	2	33
5	3	54
10	4	96

A best-versus-worst ratio alone does not predict this once several strong outcomes compete for mass.

Interpolation audit. 800 held-out off-grid points:

region	RMSE (17^3)	order
smooth	2.5×10^{-3}	1.58
near a kink	9.7×10^{-3}	1.03

So a grid is fine for exploring, but survival decisions near $m = 0$ need a real solve. The margin m is useful twice: as the criterion, and as a warning flag.

Part 5: Estimator Error Analysis

Part 5: Objective

Parts 1 to 4 assume the update knows p . It does not. It has G samples, and it needs $p^{-\alpha}$.

This is the paper's own stated limitation, and it is a pure **error-analysis** problem: bias against variance, with a knob (ε) that trades one for the other exactly like a step size trades truncation against round-off.

We compare the paper's clipped rule against three alternatives, and ask which trade the *learning dynamics* actually notice.

one group of $G = 16$ draws:

A A B A C B A D B A C B A C B A

counts n_i : $n_A=7$ $n_B=5$ $n_C=3$ $n_D=1$ $n_E=0$

weight $\omega(n_i)$: 2.3 3.2 5.3 16 **1000**

one outcome missed,
so $\omega = \varepsilon^{-1}$

Without the clip that entry is ∞ . With it, one unlucky group hands a rare outcome a weight 60 times larger than anything else in the same group.

Part 5: Numerical Methods (1/2)

Truncation error via Taylor expansion. Expand $f(x) = x^{-\alpha}$ about p with the random step $\hat{p} - p$, take expectations, and use the binomial central moments ($\mathbb{E}[\hat{p}] = p$, $\text{Var} = pq/G$):

$$\mathbb{E}[f(\hat{p})] = f(p) + \frac{1}{2}f''(p)\frac{pq}{G} + O(G^{-2}),$$

which gives the relative bias and standard deviation

$$b = \frac{\alpha(\alpha+1)(1-p)}{2Gp}, \quad s = \alpha\sqrt{\frac{1-p}{Gp}}.$$

The small parameter is Gp , the expected number of hits, not G .

Golden-section search. $\text{MSE}(\varepsilon)$ falls then rises, with kinks at every $\varepsilon = n/G$, so a derivative-free bracketing minimiser is the right tool. It shrinks the bracket by $1/\phi = 0.618$ per evaluation.

Richardson extrapolation. Equation above is an expansion in $1/G$ with G -independent coefficients, so the standard combination applies. Splitting the group into halves of size $M = G/2$:

$$\omega_R = 2f\left(\frac{n_A+n_B}{G}\right) - \frac{1}{2}\left[f\left(\frac{n_A}{M}\right) + f\left(\frac{n_B}{M}\right)\right]$$

raises the bias order from 1 to 2. Because $\hat{p}_A + \hat{p}_B = 2\hat{p}$, the first-order noise term is unchanged, so the variance is free.

Part 5: Numerical Methods (2/2)

Eigenvalue decomposition: power method and QR. A moving average over training steps turns the first-order relaxation of Part 2 into a **second-order** system. With $\kappa = \beta/h$,

$$x'' + \kappa x' + \kappa \alpha \|r\|_{1/\alpha} C x = 0 \iff s^2 + \kappa s + \kappa \lambda_j = 0.$$

The λ_j need the eigenvalues of C , which we compute with the **power method** (with Hotelling deflation) and the **QR algorithm**, both written out and checked against a library.

ODE stability, again. The quadratic rings when $\kappa < 4\lambda$ and decays fastest at $\kappa = 4\lambda$.

Round-off error. The closed form derived for $\alpha = 1$,

$$w_G(p) = \frac{1 - (1 - p)^G}{p},$$

cancels catastrophically for small p . Evaluating it as $-\expm1(G \log1p(-p))/p$ keeps the digits:

p	direct	guarded
10^{-11}	8.3×10^{-8}	2.1×10^{-15}
10^{-13}	3.1×10^{-4}	2.3×10^{-15}

Two more round-off lessons we hit for real: Gram-Schmidt QR silently breaks on the singular C , and the power method must stop on the residual $\|Ax - \lambda x\|$, not on λ , because the Rayleigh quotient is accurate to the *square* of the vector error.

Part 5: Results and Analysis

An exact optimum for the clip. Differentiating $\text{MSE}(\varepsilon)$ piecewise, the sign of the derivative is the sign of $p^{-\alpha} - \varepsilon^{-\alpha}$, so

$$\varepsilon^* = p$$

exactly, for every α , G and p . Golden-section search returns $\varepsilon^*/p = 1 \pm 7 \times 10^{-9}$, which is the $\sqrt{\epsilon_{\text{mach}}}$ floor of any derivative-free minimiser.

Since ε is one global constant but ε^* is per-outcome, **no single clip suits a majority and a minority outcome at once**. That is an honest explanation of why the paper's own ε ablation has no clear winner.

Richardson works, but must be guarded. Bias order $1.03 \rightarrow 2.10$, variance unchanged. Naive Richardson returns a **negative** weight for a singleton (-472 at $G = 16$), which is a sign-flipped reward.

The result we did not expect. The dynamics do not see $\mathbb{E}[\omega(n)]$. They see a size-biased version, $w_G(p) = \mathbb{E}[\omega(1 + B)]$ with $B \sim \text{Bin}(G - 1, p)$. Expanding it,

$$w_G(p)p^\alpha - 1 = \frac{\alpha(\alpha - 1)(1 - p)}{2Gp} + O(G^{-2}),$$

which **vanishes at $\alpha = 1$** , the paper's own exponent. In fact there it is exact: $w_G(p) = \frac{1 - (1 - p)^G}{p}$.

So correcting the estimator's bias *destroys* that cancellation. Training ℓ_1 to the ideal p^* , $K = 5$:

G	clip (paper)	guarded Richardson
8	0.2439	0.2172
16	0.1092	0.0703
64	0.0016	0.0294

The better estimator trains worse once G is large.

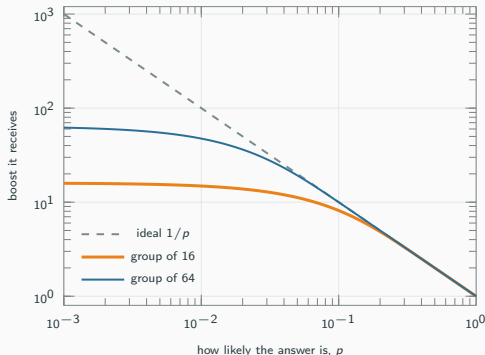
Overall Findings

Comparative Analysis

Question	The paper	What we added
Does expected return collapse onto one answer?	Proved it	Made it precise: with an exact tie nothing moves at all (10^{-16}). The random sampling is what picks a winner.
Does IPS give share $\propto$ reward?	Proved it	Confirmed to machine precision, and extended to any dial setting: share $\propto r^{1/\alpha}$.
Do the group size G and clip ε matter?	Tried a few values (their Table 4)	Derived the exact rule: the weaker answer survives only if $\alpha > \ln \rho / \ln \min\{G, 1/\varepsilon\}$. Checked on 19,125 settings with no exceptions.
Does clipping cause a bias-variance trade-off?	Named it as a limitation	Measured it exactly. The best clip is $\varepsilon = p$, and we compared three other rules.

We did not rerun the paper's language-model or molecule experiments; they need far more computing than this course. We reproduced the *mechanism*, in a small model where nothing else can explain the result.

Cross-Cutting Result: The Finite-Group Ceiling



One result that every part runs into.

The ideal boost $1/p$ keeps growing as an answer gets rarer (dashed). A real group of G samples can only give a limited boost (solid lines), because a rare answer shows up in the group at most once:

$$\text{boost} \leq \min\{G, 1/\epsilon\}^\alpha.$$

Each part meets this cap from a different side:

- Part 2 finds the cap and the two-answer rule
- Part 3 with more answers, a bigger α is needed
- Part 4 checks the rule on 19,125 settings
- Part 5 only the size of the cap decides survival

At $\alpha = 1$ the solid curves are exactly $(1 - (1 - p)^G)/p$ (Part 5).

Conclusions and Limitations

Advice for anyone using this

- **Choose the group size first.** If $\min\{G, 1/\varepsilon\}$ is smaller than best reward $\div$ worst reward, no α can keep the weak answers.
- **Keep ε at or below $1/G$.** A bigger clip adds a second cap.
- **With many answers, the two-answer rule is too hopeful.** At $G = 16$ the weakest of five needs $\alpha = 0.886$, not 0.580.
- **Averaging over steps is cheap noise reduction,** like a $19\times$ bigger group at $\beta = 0.1$. With too little damping ($\beta < 4h\lambda$) training oscillates.

What we do not claim

- We studied a simple model: one choice among K answers. It is not full GRPO with its reward normalisation and PPO clipping, so our results are about the *weight rule*.
- Our α -matched correction is accurate to second order, not exact.
- Averaging per answer needs a short list of possible answers. That is true here, not for generated text.
- We show convergence in simulations from many starting points; we do not prove it for every α .

Thank you

Questions